Atlas 178: A Half-Degree Weighted K_4 Ratio

Abstract

Graph Atlas 178 is formed from two copies of K_4 sharing a common triangle, with one pendant leaf at one of the two page vertices. We prove for every graphon W of edge density $p \in [2/3, 1]$ that

$$t(H_{178}, W) \geq p^2(2p-1)(3p-2)^2,$$

its exact chromatic target. Cauchy–Schwarz on the two pages reduces the problem to a K_4 whose page vertex carries the fractional degree weight $d^{1/2}$. Rooted-link Goodman and two sharp supporting planes in the degree, its one-step average, and the rooted triangle prove the required weighted ratio. The supporting planes are certified on the complete rooted feasible region by exact rational conditional Bernstein certificates. The proof is for arbitrary graphons and uses no finite-step reduction.

1 The graph and its target

Let H_{178} have a triangle on vertices s_1, s_2, s_3, two page vertices u, v, each adjacent to all three s_i, and a leaf adjacent to u. This graph is isomorphic to NetworkX Graph Atlas 178 and has graph6 code `En}?`. A proper colouring first chooses ordered distinct colours on the spine, then a colour outside those three for each page, and finally a colour different from the colour of u. Hence

$$P_{178}(z) = z(z-1)^2(z-2)(z-3)^2.$$

Writing

$$c_3 = 2p - 1, \qquad c_4 = 3p - 2,$$

gives

$$(1-p)^6 P_{178}\left(\frac{1}{1-p}\right) = p^2 c_3 c_4^2. \tag{1}$$

The required density interval is $p \geq 2/3$.

2 Rooted data

Write T_W for the integral operator of W, and put

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = (T_W \mathbf{1})(x),$$
$$a(x) = (T_W d)(x), \qquad \tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z). \tag{2}$$

Let

$$T = \int_\Omega \tau(x)\,\mathrm{d}\mu(x) = t(K_3, W), \qquad I = \int_\Omega d(x)^{1/2}\tau(x)\,\mathrm{d}\mu(x).$$

The standard rooted Goodman argument gives

$$T \geq p(2p-1) = pc_3. \tag{3}$$

We use the following complete scalar region. It is repeated here so the supporting-plane certificates are self-contained.

Lemma 2.1 (Rooted feasible region)**.** *For almost every x, the tuple $(d, a, t) = (d(x), a(x), \tau(x))$ satisfies*

$$\begin{gathered} 0 \leq d \leq 1, \qquad \max\{0, d+p-1\} \leq a \leq d, \\ \max\{0, 2a-p, a-d(1-d)\} \leq t \leq \min\{a, d^2\}. \end{gathered} \tag{4}$$

Moreover,

$$\int d = p, \qquad \int a = \int d^2, \qquad \int t = T. \tag{5}$$

Proof. Deleting factors gives $a \leq d$, $t \leq a$, and $t \leq d^2$. Writing $U = 1 - W$ gives

$$a(x) = p - \int U(x,y)d(y)\,\mathrm{d}\mu(y) \geq d(x) + p - 1.$$

The two inequalities

$$W(x,y)W(x,z) \geq W(x,y) + W(x,z) - 1$$

and

$$W(x,z)W(y,z) \geq W(x,z) + W(y,z) - 1$$

give, after multiplication by the remaining edge factor and integration, $t \geq 2a-p$ and $t \geq a-d(1-d)$, respectively. Nonnegativity supplies the remaining bounds. The identities in (5) are symmetry and Fubini. □

3 Two half-degree supporting planes

The next lemma is the only scalar input particular to this proof.

Lemma 3.1 (Half-degree rooted planes)**.** *Let $p \in [2/3, 1]$, set $r = \sqrt{p}$, and let (d, a, t) satisfy (4). Then*

$$(2\sqrt{d} - r)t \geq prc_3 + \frac{3}{2}r(4p-1)(d-p) + \frac{3}{2}r(a-d^2), \tag{6}$$

$$\frac{t}{\sqrt{d}}\big(2t - d^2 - c_4 d\big) \geq 3prc_3(d-p) + 2rc_3(a-d^2). \tag{7}$$

In (7), the first term is defined to be zero at $d = 0$; feasibility then forces $a = t = 0$.

The exact proof of Lemma 3.1 is given in Section 5. We first record its two consequences.

Lemma 3.2 (Half-degree triangle moment)**.** *For every graphon of density $p \in [2/3, 1]$,*

$$I^2 \geq p^2 c_3 T. \tag{8}$$

Proof. Integrating (6) and using (5) gives

$$2I - rT \geq prc_3.$$

Thus

$$I \geq \frac{\sqrt{p}}{2}(T + pc_3).$$

Together with $T \geq pc_3$, arithmetic–geometric mean yields

$$I \geq \sqrt{p}\sqrt{Tpc_3} = p\sqrt{c_3 T}.$$

Squaring proves (8). Equivalently, the exact surplus before the last step is

$$\frac{p}{4}(T + pc_3)^2 - p^2 c_3 T = \frac{p}{4}(T - pc_3)^2.$$

□

Let $\kappa_4(x)$ denote the rooted K_4 density at x, and put

$$I_4 = \int_\Omega d(x)^{1/2}\kappa_4(x)\,\mathrm{d}\mu(x).$$

Lemma 3.3 (Half-degree adjacent-clique ratio)**.** *For every graphon of density $p \in [2/3, 1]$,*

$$I_4 \geq c_4 I. \tag{9}$$

Proof. If $d(x) > 0$, use the neighbour probability measure

$$\mathrm{d}\nu_x(y) = \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu(y).$$

The edge density of the link graphon under ν_x is

$$z_x = \frac{\tau(x)}{d(x)^2},$$

whereas $d(x)^{-3}\kappa_4(x)$ is its triangle density. Goodman inside the link therefore gives

$$\kappa_4(x) \geq d(x)^3 z_x(2z_x - 1) = \frac{2\tau(x)^2}{d(x)} - d(x)\tau(x). \tag{10}$$

The assertion is trivial on $\{d = 0\}$. Multiply (10) by $\sqrt{d}$, integrate, and then integrate (7). The mean-zero correction terms disappear by (5), leaving

$$I_4 \geq 2\int \frac{\tau^2}{\sqrt{d}} - \int d^{3/2}\tau \geq c_4 \int \sqrt{d}\,\tau = c_4 I.$$

□

4 The page compression and the theorem

For a spine triple $\boldsymbol{x} = (x_1, x_2, x_3)$, write

$$A(\boldsymbol{x}) = W(x_1, x_2)W(x_1, x_3)W(x_2, x_3)$$

and, for $s \geq 0$,

$$K_s(\boldsymbol{x}) = \int_\Omega d(y)^s \prod_{i=1}^{3} W(y, x_i)\, \mathrm{d}\mu(y).$$

Integrating the pendant leaf first gives the exact identity

$$t(H_{178}, W) = \int_{\Omega^3} A(\boldsymbol{x})K_1(\boldsymbol{x})K_0(\boldsymbol{x})\, \mathrm{d}\mu^3. \tag{11}$$

Cauchy–Schwarz in the page variable gives

$$K_{1/2}(\boldsymbol{x})^2 \leq K_0(\boldsymbol{x})K_1(\boldsymbol{x}).$$

Another weighted Cauchy–Schwarz inequality, now on the spine triple, gives

$$t(H_{178}, W) \geq \int AK_{1/2}^2 \geq \frac{(\int AK_{1/2})^2}{\int A} = \frac{I_4^2}{T}. \tag{12}$$

There is no zero-denominator issue: (3) gives $T \geq 2/9$ on the required interval.

Theorem 4.1 (Atlas 178)**.** *For every graphon W with edge density $p \in [2/3, 1]$,*

$$t(H_{178}, W) \geq p^2(2p-1)(3p-2)^2.$$

Proof. Combine (12), Lemma 3.3, and Lemma 3.2:

$$t(H_{178}, W) \geq \frac{I_4^2}{T} \geq c_4^2 \frac{I^2}{T} \geq p^2 c_3 c_4^2.$$

By (1), the last expression is the exact chromatic target. □

5 Exact certificate for the rooted planes

We prove Lemma 3.1 entirely by rational polynomial certificates. Put

$$r = \frac{4+X}{5}, \qquad p = r^2, \qquad \sqrt{d} = D, \qquad (X, D) \in [0, 1]^2.$$

Thus the certificate covers $p \in [16/25, 1]$, slightly more than the required interval. For fixed p, d, the feasible polygon in the (a, t)-plane is cut out by the eight nonnegative slacks

$$a,\ d-a,\ a-d-p+1,\ t,\ t-2a+p,\ t-a+d(1-d),\ a-t,\ d^2-t. \tag{C.1}$$

Its possible vertices are

name	a	t
A	d	d^2
$B0$	$p/2$	0
$BL1$	d^2-d+p	$p-2d+2d^2$
$BL2$	$d+p-1$	$2d+p-2$
$BU1$	$(d^2+p)/2$	d^2
$BU2$	d	$2d-p$
BUp	p	p
$C0$	$d(1-d)$	0
CL	$d+p-1$	d^2+p-1

Infeasible entries are discarded by (C.1).

For (6), subtract the right side from the left. This is affine in a, t, so it suffices to check the listed feasible vertices. At $BL1$, the residual contains the square $(5D-X-4)^2$; its quotient is certified. At $BL2$, one constraint is $-(D^2-1)^2 \geq 0$, so $D=1$, where the residual factors as

$$\frac{(X-1)^2(X+4)(2X+13)(2X+23)}{3125}.$$

The exact conditional Bernstein trees are:

face	visited	accepted	infeasible
A	115	29	29
$B0$	7	0	4
$BL1$	5	2	1
$BU1$	41	16	5
$BU2$	39	10	10
BUp	15	4	4
$C0$	13	6	1
CL	265	132	1
total	500	199	55

The only two equality-corner patches are also rational. At A, write $u = 1-D, v = 1-X$. Feasibility gives

$$\frac{9}{25}v \leq 1-p \leq (1-d)^2 \leq 4u^2, \qquad v \leq \frac{100}{9}u^2.$$

The leading residual $17u^2 - v/5$ is therefore at least $133u^2/9$. On $u, v \leq 1/64$, the absolute remainder is at most

$$\frac{6332030460179381}{4057816381784064}\,u^2 < \frac{133}{9}u^2.$$

At CL, the leading quadratic is

$$17u^2 - 6uv + \frac{3}{5}v^2.$$

After subtracting $(u^2+v^2)/50$, its determinant is $2121/2500$; on $u, v \leq 1/2048$, the remainder is at most

$$\frac{697663143033}{53687091200000}(u^2+v^2) < \frac{1}{50}(u^2+v^2).$$

For (7), multiply its residual by D. The resulting polynomial is

$$Q = 2t^2 - (d^2 + c_4 d)t - 3prc_3D(d-p) - 2rc_3D(a-d^2). \tag{C.2}$$

At $D = 0$, feasibility gives $a = t = 0$, and the unmultiplied inequality is immediate. For $D > 0$, $Q \geq 0$ is equivalent to (7). The coefficient of a in (C.2) is nonpositive. Thus, for fixed d, t, the minimum occurs on one of

$$a = d, \qquad a = \frac{t+p}{2}, \qquad a = t + d(1-d).$$

The first face collapses to A. On the other two faces, Q is a convex quadratic in t, so its minimum occurs at a listed endpoint or at its stationary point. This proves that the nine vertex checks and two stationary checks below cover the whole feasible region.

face	visited	accepted	infeasible
A	67	22	12
$B0$	5	0	3
$BL1$	5	2	1
$BU1$	11	3	3
$BU2$	25	6	7
BUp	1	1	0
$C0$	7	4	0
CL	277	138	1
B-stationary	7	0	4
C-stationary	1	1	0
total	406	177	31

Again, $BL1$ and the C-stationary residual contain $(5D - X - 4)^2$, and $BL2$ collapses to $D = 1$. Its remaining boundary factor is

$$\frac{(X-1)^2(X+4)(X+9)(6X^3+102X^2+503X+514)}{78125}.$$

At the A corner, the leading quadratic is

$$18u^2 - \frac{52}{5}uv + \frac{42}{25}v^2.$$

After subtracting $(u^2+v^2)/10$, its determinant is $621/500$; on $u, v \leq 1/1024$, the remainder coefficient is

$$\frac{62190435867197668803}{900719925474099200000} < \frac{1}{10}.$$

The CL face has a thin equality tube, which is treated directly instead of forcing excessive bisection. Put $z = r - D$ and $q = 1 - r^2$. Its exact residual is

$$\begin{aligned} Q_{CL} &= \big(5r^6 + 22r^4 - 7r^2 - 2\big)z^2 + 2r(r^2+3)q^2 z + 2q^4 + z^3 R(r,z), \\ R(r,z) &= -r(30r^4 + 31r^2 - 10) + (45r^4 + 23r^2 - 3)z \\ &\quad - 2r(21r^2+5)z^2 + (25r^2+2)z^3 - 8rz^4 + z^5. \end{aligned}$$

For $9/10 \leq r \leq 1$, the first three terms are at least $\frac{3}{5}(z^2+q^4)$; the two principal minors are certified by positive one-variable Bernstein coefficients. If $|z| \leq 1/100$, then

$$|z^3 R(r,z)| \leq \frac{516552270801}{10^{12}} z^2 < \frac{3}{5} z^2.$$

Thirty Bernstein boxes lie wholly inside this proved tube. All other boxes are settled by ordinary nonnegative Bernstein coefficients or by an infeasible constraint.

Every subdivision, integer Bernstein tensor, corner coefficient, graph identity, and target factorization was reproduced in exact rational arithmetic.

6 Scope

All integrations above are over an arbitrary probability space and an arbitrary measurable symmetric kernel $0 \le W \le 1$. The proof uses Tonelli–Fubini, Cauchy–Schwarz, Goodman in a normalized link, the rooted feasible inequalities proved in Lemma 2.1, and the exact finite polynomial certificates in Section 5. It does not assume that W is finite-step, regular, or bounded away from zero. No limiting division at $d = 0$ is used.

7 What the Lean formalization actually proves

Theorem 4.1 is formalized in full, on $p \in [2/3, 1]$. The development is `lean/Taeyoung/Methods/Atlas178/`, the catalogue theorem is `satisfiesLowerBound_178`, and every declaration depends on exactly `[propext, Classical.choice, Quot.sound]`. Atlas 178 is `verified`.

Sections 1, 2 and 4 are followed as written. Section 5 is not: none of it is used.

7.1 Both planes are exact duals, not subdivision trees

The two conditional Bernstein trees of 500 and 406 boxes, the nine-vertex enumeration, the two equality-corner patches, and the Q_{CL} tube are all replaced by three polynomial identities. Writing the three feasibility slacks of (4) as

$$g_5 = t - 2a + p, \qquad g_6 = t - a + d - d^2, \qquad g_8 = d^2 - t,$$

and putting res_1 for twice the residual of (6) and res_2 for D times the residual of (7), the identities are

$$\begin{aligned}
\mathrm{res}_1 &= \tfrac{3}{2} r\, g_5 + \big(\tfrac{7}{2} r - 4D\big) g_8 + \Phi_U && \text{on } 8D \le 7r,\\
\mathrm{res}_1 &= (5r - 4D) g_5 + (8D - 7r) g_6 + 2(D - r)^2 L && \text{on } 8D \ge 7r,\\
\mathrm{res}_2 &= 2Dr c_3\, g_6 + 2(t - \theta)^2 + \tfrac{D}{8}(D - r)^2 M, &&
\end{aligned}$$

with

$$\theta = \tfrac{1}{4} D(D + r)(D^2 - Dr + 4r^2 - 2), \qquad L = 4D^3 + 6D^2 r + 8Dr^2 - 4D + 4r^3 - 3r,$$

$$\Phi_U = \tfrac{1}{2}\big(8D^5 - D^4 r - 24D^2 r^3 + 6D^2 r + 16r^5 - 5r^3\big),$$

$$M = -D^5 - 2D^4 r - 9D^3 r^2 + 4D^3 + 40D^2 r^3 - 20D^2 r + 80Dr^4 - 32Dr^2 - 4D + 48r^5 - 24r^3.$$

Every multiplier is nonnegative on its region: $5r \ge 4 \ge 4D$ because $r \ge 4/5$, and $7r/2 \ge 4D$ is the case hypothesis. So the first plane needs two cases and the second none. The tangency at the balanced tripartite graphon appears in both as an explicit $(D - r)^2$.

7.2 The three leftover factors

$L \geq 0$ on $8D \geq 7r$ is one `nlinarith`. $M \geq 0$ on the whole box is the four-term grouping

$$M = 24r^3c_3 + D(80r^4 - 41r^2 - 7) + 20D^2rc_3 + D\big(4D^2 + 9r^2 + 3 - D^4 - 2D^3r - 9D^2r^2\big).$$

$\Phi_U \geq 0$ mixes degrees 5 and 3, so its degree-3 part is first traded against its degree-5 part using $25r^2 \geq 16$; what is left is homogeneous, and in the band coordinate $7r - 8D$ it has positive coefficients throughout:

$$32\,\Phi_U = H + r(25r^2 - 16)(5r^2 - 6D^2), \qquad H = 128D^5 - 16D^4r - 234D^2r^3 + 131r^5,$$

$$256\,H = 2078r^5 + 2471(7r - 8D)r^4 + (7r - 8D)^2\big(289r^3 + 1064r^2D + 832rD^2 + 512D^3\big).$$

No box subdivision, and no Bernstein coefficient, occurs anywhere in the formalization. The certificate interval is the same: $r \geq 4/5$ is $p \geq 16/25$, the note's $X \geq 0$.

7.3 The quotient at $d = 0$

The second plane is used in the divided form of (7), since that is what has mean-zero corrections. Lean's $x/0 = 0$ is exactly the note's convention: at $d = 0$ feasibility gives $\tau \leq d^2 = 0$, so both the divided Goodman quotient $\tau(2\tau - d^2)/\sqrt{d}$ and the divided residual vanish, while the right-hand side is $-3p^2rc_3 \leq 0$. The degenerate fibre therefore needs no separate statement, only a separate two-line case.

7.4 The page identity

(11) is obtained by peeling all six coordinates and collapsing the three page coordinates in turn; the leaf integral produces $d(x_3)$ and the two page integrals produce K_1 and K_0. Both Cauchy–Schwarz steps are the same lemma, once on μ and once on the assignment measure of $\mathrm{Fin}\,3$, and the identification $\int A\,K_{1/2} = I_4$ is a single Fubini swap between the spine triple and the page variable rather than three nested ones.